

PROJECT REPORT

1. INTRODUCTION

1.1 Project Overview

The Food Ordering Behavior Analysis project aims to analyze customer food ordering patterns using Tableau. The project is based on a dataset containing 50,000 food ordering records, which includes information such as customer demographics, cities, cuisines, meal types, restaurant categories, delivery time, customer ratings, and order values.

Interactive dashboards and story visualizations were developed to identify key business trends, customer preferences, revenue distribution, and delivery performance. The insights generated from the analysis can help food delivery businesses understand customer behavior and make informed decisions to improve operational efficiency and customer satisfaction.

Purpose of the Project:

The primary purpose of this project is to analyze customer food ordering behavior and transform raw data into meaningful business insights through interactive Tableau visualizations.

The project focuses on:

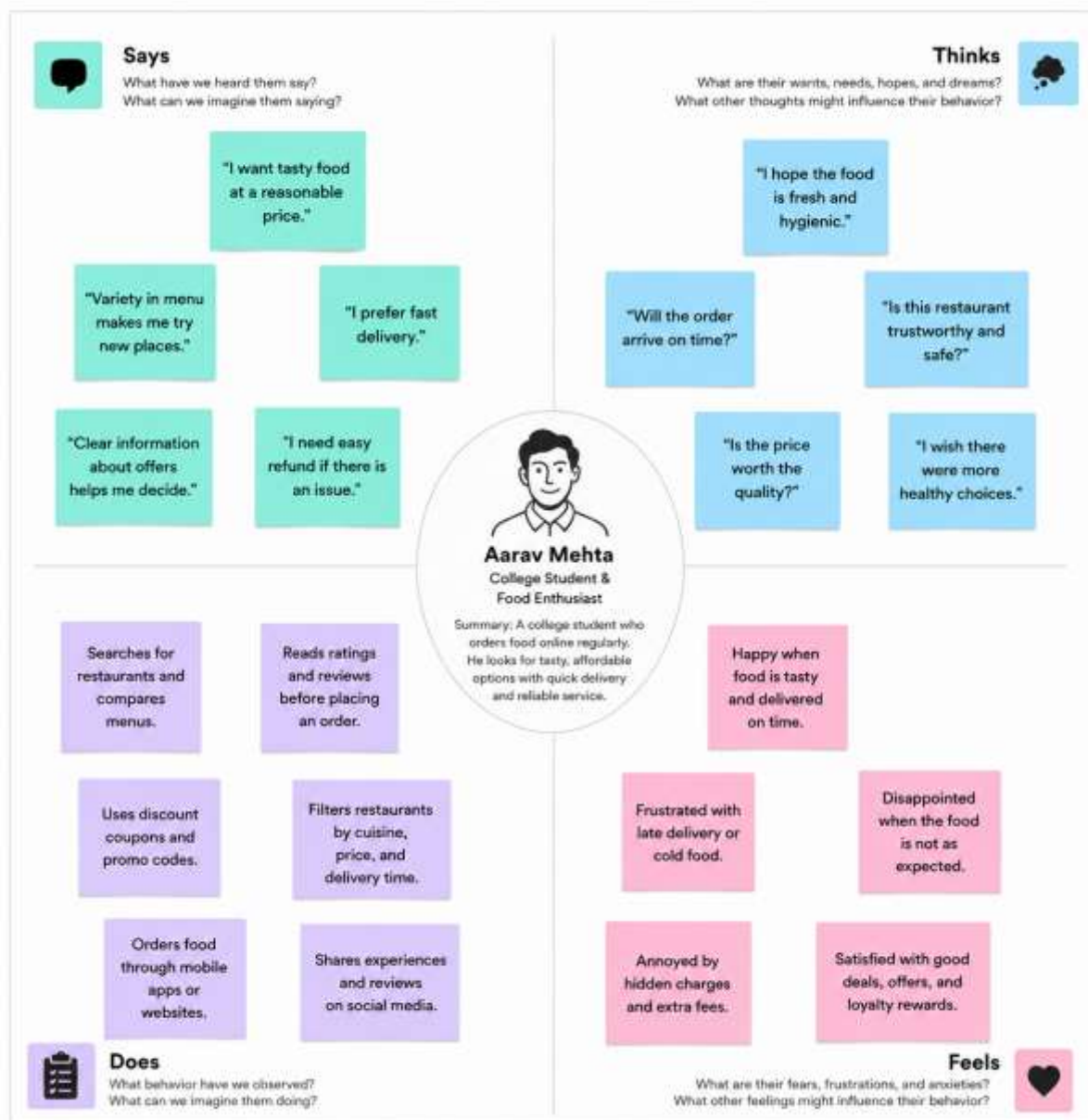
- Understanding customer ordering patterns across different cities.
- Identifying the most preferred cuisines and meal types.
- Analyzing customer demographics and ordering behavior.
- Evaluating revenue contribution across different meal categories.
- Assessing customer satisfaction through ratings.
- Monitoring delivery performance using delivery time analysis.
- Supporting data-driven decision-making for business growth and improved customer experience.

2. IDEATION PHASE

2.1 Problem Statement



2.2 Empathy Map Canvas



2.3 Brainstorming

Step-1: Team Gathering, Collaboration and Select the Problem Statement:



Brainstorming & ideation

Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits

Use this template in your own brainstorming sessions so your team can uncover insights into food ordering behaviour, preferences, and emerging consumer trends.

⌚ 90 minutes to complete
 🕒 1 hour to collaborate
 👥 2-8 people recommended

➔ Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

⌚ 10 minutes

- ☐ **Team gathering**
Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.
- ☐ **Set the goal**
Think about the problem you'll be focusing on solving in the brainstorming session.
- ☐ **Learn how to use the facilitation tools**
Use the facilitator Superpowers to run a happy and productive session.

[Open article](#) ➔

1 Define your problem statement

What specific aspect of food ordering behaviour or consumer trends do you want to explore? Frame your focus as a How Might We statement. This will guide the rest of your exploration.

⌚ 5 minutes

EXAMPLE

How might we enhance the food ordering experience to drive loyalty and healthier choices?



Key rules of brainstorming

To run an smooth and productive session

- 🗣️ Stay in topic. 💡 Encourage wild ideas.
- ⏸️ Defer judgment. 👂 Listen to others.
- 🗣️ Go for volume. 👁️ If possible, be visual.

Step-2: Brainstorm, Idea Listing and Grouping:

1

Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

TIP

Go for as many ideas as possible—quantity over quality is the goal!

Anar

Yashesh

Shravan

Thaan

Taste

Price

Place

Food

Quality

Discounts

Delivery

Fast

Delivery

Restaurant

Rating

Reviews

Variety

Healthy

Food

Hygiene

Packaging

Ali

Mohan

Anand

Suhail

Convenience

Time

Saving

Payment

Options

Brand

Trust

Customer

Service

Others

Online

Reviews

Delivery

Tracking

Availability

Loyalty

Rewards

Promotions

Order

Accuracy

2

Group Ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a name or theme label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

20 minutes

TIP

Add a customer label to sticky notes to make it easier to find, organize, and categorize important ideas as there's value you can't lose.

Price & Promotions

Food Quality

Discounts

Coupons

Loyalty

Rewards

Taste

Freshness

Hygiene

Healthy

Options

Convenience

Customer Experience

Fast

Delivery

Payment

Methods

Easy

Ordering

Order

Tracking

Restaurant

Reviews

Packaging

Customer

Support

Step-3: Idea Prioritization:

3

Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

TIP

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Support

3. Requirement Analysis

3.1 Customer Journey Map



3.2 Solution Requirements

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Google Account
FR-2	User Authentication	Login using Email and Password ,Password Recovery via Email
FR-3	Food Search	Search food items by name Filter restaurants by category, rating, and location
FR-4	Food Ordering	Add food items to cart Place online food orders Track order status
FR-5	Consumer Behaviour Analysis	Analyze customer ordering behaviour Analyze food preferences and purchasing patterns
FR-6	Food Recommendation	Recommend personalized food items Suggest popular restaurants based on user preferences

FR-7	Feedback Management	Submit customer reviews and ratings Analyze customer feedback
FR-8	Report Generation	Generate consumer behaviour reports Export reports in PDF or Excel format
FR-9	Administrator Management	Manage users, restaurants, menus, and customer orders
FR-10	Machine Learning Prediction	Predict consumer food preferences and identify ordering trends

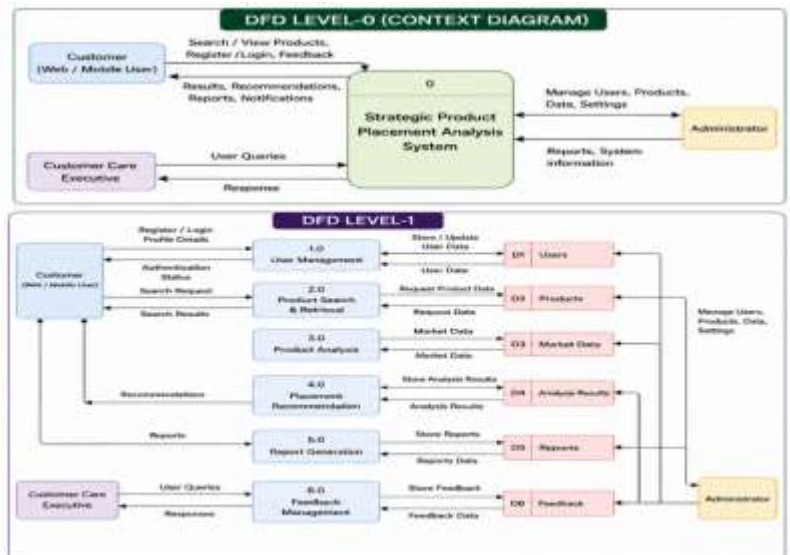
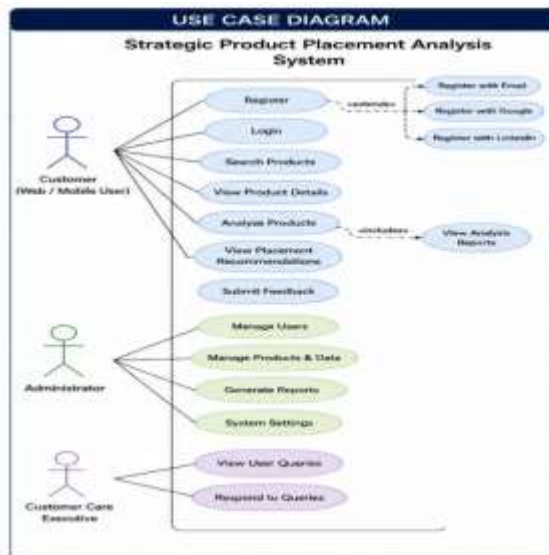
Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

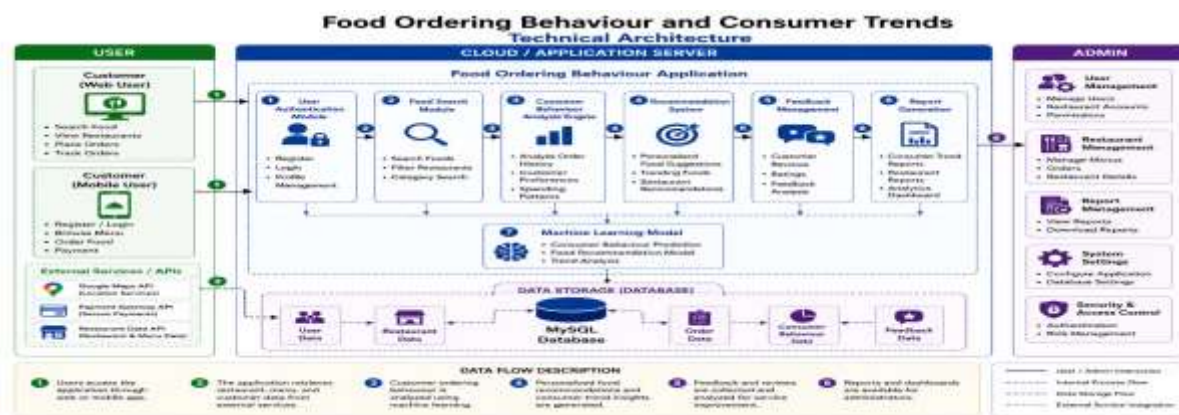
FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provides an easy-to-use and user-friendly interface for customers and administrators.
NFR-2	Security	Ensures secure authentication, password encryption, and protection of customer information.
NFR-3	Reliability	Generates accurate consumer behaviour analysis and recommendation results.
NFR-4	Performance	Provides fast food search, recommendation generation, and order processing.
NFR-5	Availability	System is available 24×7 with cloud-based hosting and database backup.
NFR-6	Scalability	Supports increasing numbers of customers, restaurants, food items, and orders.
NFR-7	Maintainability	Allows easy updates to menus, datasets, recommendation models, and application features.
NFR-8	Compatibility	Supports major web browsers and mobile devices for seamless access.

3.3 Data Flow Diagram

Example: Strategic Product Placement Analysis System [\(Simplified\)](#)



3.4 Technology Stack



4. PROJECT DESIGN

4.1 Problem Solution Fit

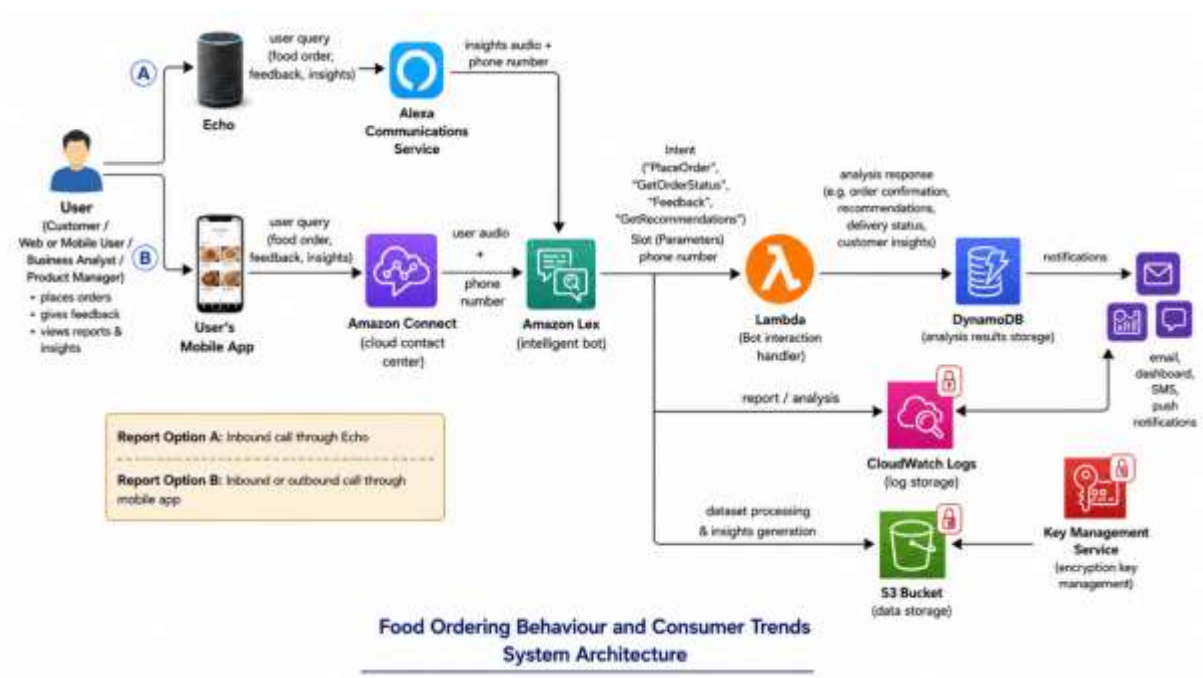
Define CS, AS into CC	1. CUSTOMER SEGMENT(S) CS Who is your customer? - Online food ordering customers - Students - Working professionals - Families - Restaurant owners - Food delivery partners - Food enthusiasts	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices of solutions? (i.e., spending power, budget, no cash, network connectivity, available devices) - High delivery charges - Long delivery time - Fake or misleading reviews - Limited payment options - Poor recommendation quality - Menu availability issues - Internet connectivity problems	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What price & core do these solutions have? (i.e., pen and paper is an alternative to digital ordering) - Swiggy - Zomato - Uber Eats - Restaurant websites - Phone ordering - Food delivery apps	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (or problems) do you address for your customer? There could be more than one, explore different sides. - Find quality food quickly - Compare restaurants and prices - Order food online - Receive personalized recommendations - Track food orders - Save time while ordering - Enjoy discounts and offers	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? (i.e., customers have to do it because of the change in regulations) - Too many restaurant choices - Lack of personalized recommendations - Inconsistent food quality - High delivery costs - Delayed deliveries - Poor customer experience - Lack of consumer behaviour insights	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? (i.e., directly related: find the right side panel, install, calculate usage and benefits, indirectly associated: customers spend less time on job-related work (i.e., time spent) - Search restaurants - Compare ratings and reviews - Compare prices - Check delivery time - Apply coupons - Order favorite foods - Leave ratings and reviews - Reorder previous meals	Focus on J&P, RC into BE, differentiate
Identify strong TR, in EM	3. TRIGGERS TR What triggers customers to act? (i.e., seeing their neighbour's building prior panels, meeting about a more efficient solution in the news) - Hunger or meal time - Attractive discounts - Festival offers - Weekend dining - Busy work schedule - Recommendation from friends - Caving for a particular cuisine	10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution first. Be in the canvas, and check how much it fits really. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour. Food Ordering Behaviour and Consumer Trends System - Collects customer ordering behaviour data. - Analyzes food preferences using Machine Learning. - Provides personalized food recommendations. - Predicts consumer ordering trends. - Helps restaurants understand customer preferences. - Improves customer satisfaction and ordering experience. - Generates consumer behaviour analytics and reports.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7 - Food delivery mobile apps - Restaurant websites - Social media - Google Search - Email notifications - Push notifications 8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development. - Restaurant visits - Word-of-mouth recommendations - Printed advertisements - Food festivals - Friends and family	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? (i.e., feel, become in-control, in-control - use in your communication strategy & design) Before: Hungry, confused by many choices, unsure about food quality, looking for discounts After: Satisfied, happy, saved time, enjoyed good food, likely to order again			

4.2 Proposed Solution

S.No	Parameter	Description
1	Problem Statement	Customers often face difficulty in choosing suitable food items and restaurants due to numerous options, varying prices, ratings, and changing consumer preferences.
2	Idea / Solution Description	The system analyzes food ordering behaviour, customer preferences, purchase history, and restaurant data to recommend personalized food items and identify consumer trends using machine learning techniques.
3	Novelty / Uniqueness	Uses machine learning algorithms to analyze customer behaviour and provide personalized food recommendations, trend analysis, and intelligent decision support for restaurants and customers.
4	Social Impact / Customer Satisfaction	Improves customer satisfaction by providing accurate food recommendations, reducing search time, and helping restaurants understand consumer preferences for better service.

5	Business Model (Revenue Model)	Revenue is generated through premium restaurant subscriptions, featured restaurant promotions, advertising, and data analytics services for restaurant owners.
6	Scalability of the Solution	The system can be scaled to support millions of customers, multiple restaurants, cloud-based deployment, real-time recommendations, and expanding food delivery platforms.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As an analyst, I can collect food ordering data from various sources for analysis.	2	High	S. Shabaz Banu, S. Fathima Mahek

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Preparation	USN-2	As an analyst, I can clean the dataset by removing missing and duplicate values.	3	High	S. Shabaz Banu, S. Fathima Mahek
Sprint-1	Data Preparation	USN-3	As an analyst, I can create calculated fields such as total orders, revenue, and average order value.	3	High	S. Shabaz Banu, S. Fathima Mahek
Sprint-1	Data Preparation	USN-4	As an analyst, I can categorize food items into consistent categories (veg, non-veg, beverages, etc.).	3	Medium	S. Shabaz Banu, S. Fathima Mahek
Sprint-2	Data Visualization	USN-5	As a user, I can view sales performance using bar charts.	2	High	S. Shabaz Banu, S. Fathima Mahek
Sprint-2	Data Visualization	USN-6	As a user, I can analyze category-wise food distribution using pie charts.	2	Medium	S. Shabaz Banu, S. Fathima Mahek
Sprint-2	Data Visualization	USN-7	As a user, I can identify order trends over time using line charts.	2	High	S. Shabaz Banu, S. Fathima Mahek
Sprint-2	Data Visualization	USN-8	As a user, I can compare restaurant or location-wise performance using maps.	4	High	S. Shabaz Banu, S. Fathima Mahek
Sprint-2	Dashboard	USN-9	As a user, I can view an interactive food ordering dashboard with KPIs, filters, and insights.	5	High	S. Shabaz Banu, S. Fathima Mahek

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2	Story	USN-10	As a user, I can view a Tableau Story summarizing key insights and business recommendations.	5	High	S. Shabaz Banu, S. Fathima Mahek

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	12	6 Days	29 Jun 2026	04 Jul 2026	12	04 Jul 2026
Sprint-2	20	6 Days	05 Jul 2026	10 Jul 2026	20	10 Jul 2026

Velocity:

The project consists of 2 sprints, each with a duration of 6 days. The total story points completed are 32, resulting in a team velocity of 16 story points per sprint.

Average Velocity (AV) per day = Velocity ÷ Sprint Duration

AV = 16 ÷ 6 = 2.67 story points per day.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

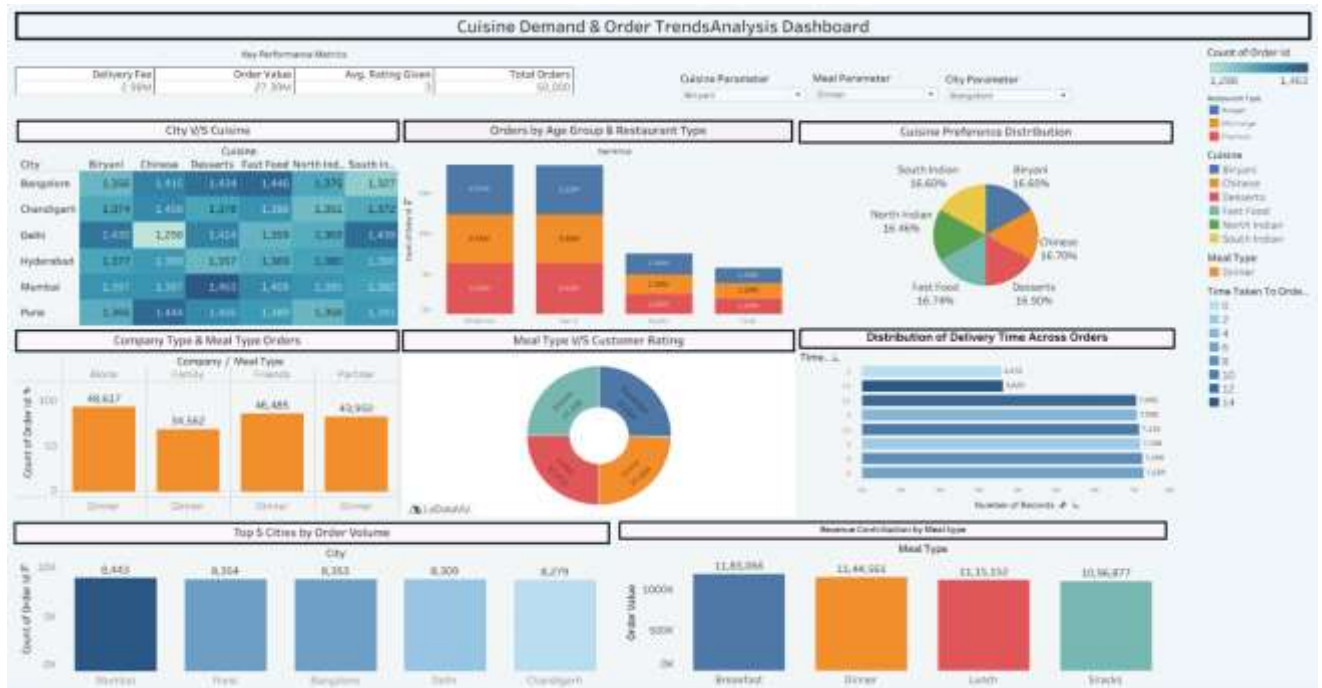
Model Performance Testing:

S.No	Parameter	Screenshot / Values
1	Data Rendered	Successfully rendered 50,000 records. Dashboard KPIs, filters and visualizations

		display correctly without data loss.
2	Data Preprocessing	Verified data types, handled missing values, removed duplicates, validated categorical and numerical fields, and created calculated fields.
3	Utilization of Filters	3 Filters Used: • Cuisine • Meal Type • City
4	Calculated Fields Used	Delivery Fee, Order Value, Average Rating Given, Total Orders
5	Dashboard Design	No. of Visualizations: 8 1. Heatmap – City vs Cuisine 2. Stacked Bar – Orders by Age Group & Restaurant Type 3. Pie Chart – Cuisine Preference Distribution 4. Bar Chart – Company Type vs Meal Type Orders 5. Doughnut Chart – Meal Type vs Customer Rating 6. Horizontal Bar – Delivery Time Distribution 7. Bar Chart – Top 5 Cities by Order Volume 8. Bar Chart – Revenue Contribution by Meal Type.
6	Story Design	The dashboard explains cuisine demand, customer preferences, age-group ordering patterns, delivery time, city-wise demand, meal-wise revenue and business insights using interactive filters.

7. RESULTS

7.1 Output Screenshots



DAHBOARD:

https://public.tableau.com/views/FoodorderingDashboard_17831614663950/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

STORY:

https://public.tableau.com/views/FoodorderingStory_17831615318620/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

ADVANTAGES & DISADVANTAGES:

Advantages:

- Provides valuable insights into customer ordering behavior and preferences.
- Helps identify the most popular cuisines, meal types, and cities with high order volumes.
- Supports data-driven business decisions through interactive Tableau dashboards.
- Enables businesses to understand customer demographics and purchasing patterns.

- Assists in improving marketing strategies by targeting high-demand customer segments.
- Facilitates performance monitoring using KPIs such as total orders, revenue, customer ratings, and delivery time.
- Interactive filters allow users to analyze data based on different categories for better decision-making.

Disadvantages:

- The analysis is limited to the available dataset and may not represent real-time customer behavior.
- External factors such as weather, festivals, promotions, and seasonal demand are not considered.
- Customer preferences may change over time, requiring regular dataset updates.
- The dashboard provides descriptive insights but does not predict future customer behavior.
- Any incorrect or incomplete data can affect the accuracy of the analysis and visualizations.

2. CONCLUSION

The Food Ordering Behavior Analysis project successfully analyzes customer ordering patterns, cuisine preferences, meal types, and delivery performance using Tableau. The dashboard provides meaningful insights into customer behavior through interactive visualizations and KPIs. The analysis identifies key business trends, including high-demand cities, popular cuisines, revenue-generating meal types, and the dominant customer age groups.

Overall, this project demonstrates how data visualization can support informed decision-making, improve operational efficiency, and help food delivery businesses develop effective marketing and customer engagement strategies. The interactive dashboard enables stakeholders to quickly explore the data and gain actionable insights for business growth.

3. FUTURE SCOPE

- Integrate real-time food ordering data for live business monitoring.
- Apply machine learning techniques to predict customer ordering behavior and demand.
- Develop personalized food recommendation systems based on customer preferences.

- Include additional factors such as weather conditions, festivals, and promotional campaigns for deeper analysis.
- Expand the dashboard with forecasting models to estimate future sales and revenue.
- Develop a mobile-friendly dashboard for easy access by business managers.
- Integrate customer feedback and sentiment analysis to improve service quality.

11. APPENDIX

Source Code(if any)

Dataset Link:

[https://drive.google.com/file/d/1GhnkK7vumWDJJ6zhsrJXuDW9fcYnrqtr/view?usp=drive link](https://drive.google.com/file/d/1GhnkK7vumWDJJ6zhsrJXuDW9fcYnrqtr/view?usp=drive_link)

Github & Project Demo Link:

Github Link:

<https://github.com/SyedShabaz18/Food-ordering-behaviour-Consumer-Trends>

Demo Link:

<https://drive.google.com/file/d/1rjcinekBgAWPw7SZO7UySPX2DpdFN-iO/view?usp=drivesdk>